

◉ Linear Regression

→ Continuous Relation

(i) Univariate

(ii) Multivariate

◉ Logistic Regression

→ Classification

◉ Polynomial Regression

→ For non-linear relation

→ Univariate Linear Regression

$$\hat{y} = \beta_0 + \beta_1 x$$

$$w_1 = \frac{N(\sum x_j y_j) - (\sum x_j)(\sum y_j)}{N(\sum x_j^2) - (\sum x_j)^2}$$

$$RMSE = \sqrt{\sum_{i=1}^n \frac{1}{n} (\bar{y}_i - \hat{y}_j)^2}$$

Accuracy,

$$R^2 = \frac{1 - \frac{\sum (y_i - \hat{y})}{\sum (y_i - \bar{y})}}{1}$$